LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Matter in Our Surroundings

Chapter / Topic: Matter in Our Surroundings (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. K = °C + 273

SECTION A

6 marks

Choose the correct option.

Q1. In which state is the space between particles the least? **[1]**

(a) Liquid (b) Gas
(c) Same in all three (d) Solid

Q2. The boiling point of water on the Kelvin scale is: **[1]**

(a) 100 K (b) 273 K
(c) 373 K (d) 473 K

Q3. The rate of evaporation increases with an increase in: **[1]**

(a) humidity (b) wind speed
(c) the depth of the liquid (d) air pressure only

Q4. Dry ice is solid: **[1]**

(a) water (b) nitrogen
(c) ammonia (d) carbon dioxide

Q5. The change of a gas into a liquid is called: **[1]**

(a) evaporation (b) sublimation
(c) condensation (d) melting

Q6. **Assertion (A):** The temperature stays constant while ice is melting.
Reason (R): The heat supplied is used to overcome the forces of attraction between the particles (latent heat). **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. The smell of hot, sizzling food reaches you several metres away, but you have to go close to cold food to smell it. Why? **[2]**

OR

Why is ice at 273 K more effective at cooling than water at the same temperature?

Q8. Convert (a) 0 °C to kelvin, (b) 310 K to degrees Celsius. **[2]**

SECTION C

6 marks

Q9. Answer the following about diffusion: **[3]**

- **(i)** What is diffusion? **[1]**
- **(ii)** Why is diffusion faster in gases than in liquids? **[1]**
- **(iii)** Give one everyday example of diffusion. **[1]**

Q10. How can a gas be changed into a liquid? Explain with the example of LPG. **[3]**

OR

Explain: (a) a sponge can be compressed, though it is a solid; (b) naphthalene balls disappear over time; (c) we can smell perfume from across a room.

SECTION D

5 marks

Q11. Answer the following: **[5]**

- **(i)** Describe an activity to study how temperature changes when ice is heated until the water boils. **[3]**
- **(ii)** Define latent heat of fusion and latent heat of vaporisation. **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. In summer, Meera noticed that wet clothes dried much faster on a sunny, windy day than on a cloudy, still day. Her grandmother kept drinking water in an earthen pot (matka), and it stayed cool. In the evening, people sprinkled water on the hot terrace floor to cool it. **[4]**

- **(i)** Name the process common to all three observations. **[1]**
- **(ii)** Why does water in an earthen pot stay cool? **[1]**
- **(iii)** Give two reasons why the clothes dried faster on the sunny, windy day. **[2]**

~ All the best! ~